

MIS 4693: Intelligent Robotic Automation

Pre-requisites: MIS 3213, or departmental permission

Instructor:

Jonathan Ye, Ph.D.
Associate Professor, Michael F. Price College of Business
Adams Hall (Main Campus)
Course website: <https://canvas.ou.edu/>

Office Hours:

Tuesday: 12:00-2:00 pm; Or appointment by email

Monday and Wednesday: 11:00 am-12:00 pm (by TA)

Pre-requisite:

Students must complete MIS 3213 as a prerequisite to this course.

Course Objective

This course introduces students to the latest intelligent robotic automation and AI technology (e.g., ChatGPT). Particular attention is given to emerging issues associated with the building and applications of AI technology architectures to enable efficient, effective, and adaptable business operations and strategies. It also provides hands-on exercises for building intelligent robots to address emerging business needs (e.g., incorporating Chat GPT for robot building, data populating, web data scraping, invoice processing, creating intelligent bots, etc.).

In this course, we will cover four major topics:

- AI for Strategy and Business processes
- RPA
- Chat GPT
- IQ bots and cognitive automation
- Emerging technologies

Learning Outcomes

- (1) Develop bots for business tasks
- (2) Develop AI agents
- (3) Understand what tasks to automate and how to maintain AI agents
- (4) Deliver AI solutions to business needs

Course Materials

1. Textbook:

Mullakara, N., and Asokan, A. 2020. "Robotic Process Automation Projects: Build Real-World RPA Solutions Using UiPath and Automation Anywhere." Birmingham, UK: Packt Publishing Ltd.

E-copy available in the library: https://ou-primo.hosted.exlibrisgroup.com/primo-explore/fulldisplay?docid=TN_cdi_proquest_ebookcentral_EBC6212528&vid=OUNEW&context=PC

Taulli, T.: The Robotic Process Automation Handbook: A Guide to Implementing RPA Systems. Apress, Berkeley (2020).

2. Other additional materials will be made available in Canvas.

Note: I expect everyone to have read all assigned materials before class. Questions for the quizzes and final exam will largely reflect the issues addressed in the reading and associated class discussion and exercises.

Technical Requirement

You need (1) a **Windows** laptop, and (2) a wireless Internet connection in class. We will use laptops regularly in the classroom. Given the nature of this course, you should bring a well-functioning **Windows** laptop with the following software installed to each class.

1. **RPA.** We will use UiPath and Power Automate for the RPA labs. Details regarding how to download and use the tools will be available on Canvas.

Grading

Various forms of activities can be used in the course:

Final exam	28%
Group project	30% (2 group projects)
Individual programming exercises	28% (14 exercises, 2% each)
Participation	14%
Total	100%

Final Exam (28%)

This class has a take-home final exam.

Group project (30%)

We have two group projects on RPA. Each group consists of around 4 to 5 people. The group will work together on the two projects about RPA. The deliverables will be a report, bots, and an in-class presentation. Details will be available in class.

Dealing with the "Free-Loader" Problem

As a large portion of your grade is linked to teamwork, the potential for "free-loading" does exist. If it does arise, the following steps must be taken:

- Groups must directly and openly confront instances of such behavior as early as possible. Ideally, once confronted, the behavior will cease, and no further steps will be required.
- If early group efforts to confront the problem fail, team members must let me know as soon as possible that a problem exists. I will schedule a meeting with all group members to discuss the problem and its causes. Hopefully, the behavior will then cease, and no further steps are required. This meeting must take place early in the module.

There is little I can do if groups delay letting me know about a problem with only one or two weeks left in the course. What I can and will do is adjust the 'group' portion of each individual grade as explained below to account for relative individual contributions to a group's efforts.

Peer Evaluations

Peer evaluations are confidential and won't be shared with anybody. Your peer evaluation score will be calculated as the average of every team member's assessment including your own. Failure to provide peer evaluation will result in a 0 for your peer evaluation (this will significantly reduce your calculated average). The total points for the project will be adjusted based on this score. Here is how the peer evaluation score will influence your total for the project:

- 7-10 points - - no change to the final team score
- 4-6 points - - final team score will be multiplied by 0.9
- 1-3 points - - final team score will be multiplied by 0.8

Individual programming exercises (28%, 14 exercises, 2% each)

We will have 14 exercises. For each exercise, the student needs to complete the robot building and turn in a short report. Details about the reports will be available in class.

Participations (14%)

Participation is essential for this course. Therefore, a significant factor in determining grades will be the degree to which you participate in class. The participation grade is computed as follows:

Attendance and active in-class participation – 11%, Professionalism – 3%.

One participation point will be deducted for each absence, failure to participate in labs and other class activities, and deficiencies in professionalism (such as improper laptop use, talking in class, arriving late, or leaving early, etc.). Only if a student is attending a university-sponsored event or has a well-documented illness will be excused. To get a full participation score, you also need to participate in class activities and discussions very actively.

Grading will be based on a standard scale: (A=90+, B=80+, C=70+, D=60+, F=60-)

Weekly Schedule

(Note: The content of this schedule is subject to changes throughout the semester)

Week	Topic	Readings	Assessments
1	Introduction/ IS Values	how IS adds value to organizations	Form a group
	AI & Business Strategy	• How AI can affect business strategies	
2	Aligning AI strategy	• Strategy for AI, Align AI with business strategy • Case discussion: Butterfly Network with AI	
	Intro to RPA	• RPA and its applications, Install UiPath	
3	• RPA 1: Create a simple bot		Exercise 1
	• RPA variables		Exercise 2
	• RPA flow control and loops		
4	• Excel processing: Excel manipulation Exercise		Exercise 3
	File handling		Exercise 4
5	Application/browser manipulation: Screenshot and auto-commenting		Exercise 5
	Browser data input: Populating data into a browser		Exercise 6
6	Data Scraping		Exercise 7
	Unattended bots and orchestrator management		
7	Publish bots to Orchestrator		Exercise 8
	Group project working week		
8	Group project presentation		Group project 1
9	Spring Break		
10	ChatGPT		Exercise 9
	OCR / Computer vision		Exercise 10
	Document understanding- Digitize the Document		
	Invoice handling- Extracting information from scanned invoices		
11	• Cognitive automation/ Intelligent automation		
	• Digital workforce; No/low code platforms		
	Power Automate Introduction		
12	Power Automate- Twitter sentiment monitoring		Exercise 11

	Power Automate- Invoice handling	Exercise 12
13	Power Automate- AI builder	Exercise 13
	Power Automate- Copilot	Exercise 14
14	RPA Review	
	Disruptive technologies Case discussion: Lemonade and AI	
15	Group project 2 working and presentation	Group project 2
16	Final exam review	

Generative AI Policy:

I permit the use of standalone Generative AI tools like ChatGPT, Copilot, Claude, etc., for the following:

- Searching for publications on your research topic.
- Creating an outline, text, or speech for presentations and write-ups.
- Providing grammatical feedback on drafts.
- Further study on concepts outside of class.
- Troubleshooting problems with software.
- Creating initial ideas for the project.

Whenever you use Generative AI for any of these, you are required to make a declaration acknowledging how and to what extent you used it. Please include an appendix that details the prompts you used and the responses you obtained. Usage of Generative AI outside of the clearly defined scope, and without acknowledgment as required, will be considered a violation of the academic integrity policy for this course.

Academic Integrity/Misconduct:

I expect everyone to abide the university's honor code:

"On my honor, as a student at the University of Oklahoma, I affirm that I will neither give nor receive inappropriate aid in the completion of any academic exercise. I understand that it is my responsibility to comply with the Academic Misconduct Code."

OU Resources on Copyright and Piracy:

<http://www.ou.edu/content/ouit/security/copyright.html>

University Policies

Mental Health Support Services

Support is available for any student experiencing mental health issues that are impacting their academic success. Students can either be seen at the University Counseling Center (UCC) located on the second floor of Goddard Health Center or receive 24/7/365 crisis support from a licensed mental health provider through [TELUS Health](#). To schedule an appointment or receive more information about mental health resources at OU please call the UCC at 405-325-2911 or visit [University Counseling Center](#). The UCC is located at 620 Elm Ave., Room 201, Norman, OK 73019.

Title IX Resources and Reporting Requirement

The University of Oklahoma faculty are committed to creating a safe learning environment for all members of our community, free from gender and sex-based discrimination, including sexual harassment, domestic and dating violence, sexual assault, and stalking, in accordance with Title IX. There are resources available to those impacted, including: speaking with someone confidentially about your options, medical attention, counseling, reporting, academic support, and safety plans. If you have (or someone you know has) experienced any form of sex or gender-based discrimination or violence and wish to speak with someone confidentially, please contact [OU Advocates](#) (available 24/7 at 405-615-0013) or [University Counseling Center](#) (M-F 8 a.m. to 5 p.m. at 405-325-2911)

Because the University of Oklahoma is committed to the safety of you and other students, and because of our Title IX obligations, I, as well as other faculty, Graduate Assistants, and Teaching Assistants, are mandatory reporters. This means that we are obligated to report gender-based violence that has been disclosed to us to the Institutional Equity Office. This means that we are obligated to report gender-based violence that has been disclosed to us to the Institutional Equity Office. This includes disclosures that occur in: class discussion, writing assignments, discussion boards, emails and during Student/Office Hours. You may also choose to report directly to the Institutional Equity Office. After a report is filed, the Title IX Coordinator will reach out to provide resources, support, and information and the reported information will remain private. For more information regarding the University's Title IX Grievance procedures, reporting, or support measures, please visit [Institutional Equity Office](#) at 405-325-3546.

Reasonable Accommodation Policy

The University of Oklahoma (OU) is committed to the goal of achieving equal educational opportunity and full educational participation for students with disabilities. If you have already established reasonable accommodations with the Accessibility and Disability Resource Center (ADRC), please [submit your semester accommodation request through the ADRC](#) as soon as possible and contact me privately, so that we have adequate time to arrange your approved academic accommodations.

If you have not yet established services through ADRC, but have a documented disability and require accommodations, please complete [ADRC's pre-registration form](#) to begin the registration process. ADRC facilitates the interactive process that establishes reasonable accommodations for students at OU. For more information on ADRC registration procedures, please review their [Register with the ADRC](#) web page. You may also

contact them at (405)325-3852 or adrc@ou.edu, or visit www.ou.edu/adrc for more information.

Note: disabilities may include, but are not limited to, mental health, chronic health, physical, vision, hearing, learning and attention disabilities, pregnancy-related. ADRC can also support students experiencing temporary medical conditions.

Religious Observance

It is the policy of the University to excuse the absences of students that result from religious observances and to reschedule examinations and additional required classwork that may fall on religious holidays, without penalty. [\[See Faculty Handbook 3.15.2\]](#)

Adjustments for Pregnancy/Childbirth Related Issues

Should you need modifications or adjustments to your course requirements because of documented pregnancy-related or childbirth-related issues, please contact the Accessibility and Disability Resource Center at 405/325-3852 and/or the Institutional Equity Office at 405/325-3546 as soon as possible. Also, see the Institutional Equity Office [FAQ on Pregnant and Parenting Students' Rights](#) for answers to commonly asked questions.

Final Exam Preparation Period

Pre-finals week will be defined as the seven calendar days before the first day of finals. Faculty may cover new course material throughout this week. For specific provisions of the policy please refer to OU's [Final Exam Preparation Period policy](#).

Emergency Protocol

During an emergency, there are official university [procedures](#) that will maximize your safety.

Severe Weather: If you receive an OU Alert to seek refuge or hear a tornado siren that signals severe weather.

1. Look for severe weather refuge location maps located inside most OU buildings near the entrances.
2. Seek refuge inside a building. Do not leave one building to seek shelter in another building that you deem safer. If outside, get into the nearest building.
3. Go to the building's severe weather refuge location. If you do not know where that is, go to the lowest level possible and seek refuge in an innermost room. Avoid outside doors and windows.
4. Get in, Get Down, Cover Up
5. Wait for official notice to resume normal activities.

Additional [Weather Safety Information](#) is available through the Department of Campus Safety.

The University of Oklahoma Active Threat Guidance

The University of Oklahoma embraces a Run, Hide, Fight strategy for active threats on campus. This strategy is well known, widely accepted, and proven to save lives. To

receive emergency campus alerts, be sure to update your contact information and preferences in the account settings section at one.ou.edu.

RUN: Running away from the threat is usually the best option. If it is safe to run, run as far away from the threat as possible. Call 911 when you are in a safe location and let them know from which OU campus you're calling from and location of active threat.

HIDE: If running is not practical, the next best option is to hide. Lock and barricade all doors; turn off all lights; turn down your phone's volume; search for improvised weapons; hide behind solid objects and walls; and hide yourself completely and stay quiet. Remain in place until law enforcement arrives. Be patient and remain hidden.

FIGHT: If you are unable to run or hide, the last best option is to fight. Have one or more improvised weapons with you and be prepared to attack. Attack them when they are least expecting it and hit them where it hurts most: the face (specifically eyes, nose, and ears), the throat, the diaphragm (solar plexus), and the groin.

Please save OUPD's contact information in your phone.

NORMAN campus: *For non-emergencies call (405) 325-1717. For emergencies call (405) 325-1911 or dial 911.*

TULSA campus: *For non-emergencies call (918) 660-3900. For emergencies call (918) 660-3333 or dial 911.*

Fire Alarm/General Emergency [OU Fire Safety on Campus](#)

If you receive an OU Alert that there is danger inside or near the building, or the fire alarm inside the building activates:

1. *LEAVE* the building. Do not use the elevators.
2. *KNOW* at least two building exits
3. *ASSIST* those that may need help
4. *PROCEED* to the emergency assembly area
5. *ONCE safely outside, NOTIFY first responders of anyone that may still be inside building due to mobility issues.*
6. *WAIT* for official notice before attempting to re-enter the building.